

Instructor: Andy Huchala
Office: Fenton 311
Email: Ahuchala@uoregon.edu
Classroom: Straub 254

Office Hours:
see when2meet link below

Class Meetings: MTWR 3:00 –3:50 pm Straub 254

Course Materials:

- We will primarily use the online notes at cruzgodar.com/teaching/uo/253 written by Cruz Godar.
- **Textbook:** The official textbook for this course is *OpenSTAX Calculus Volume II*. An electronic edition of this text is available for free at <http://openstax.org/details/books/calculus-volume-2>
The course covers most of Chapters 5–6.
- **Calculator:** No calculators will be allowed on quizzes or exams, but you can use any calculator for homework assignments.
- **Scanning Ability:** Weekly homework will be submitted in person at the start of class on **Mondays**. If you are unable to make it to class, please upload a scan of your homework to Canvas before 3:00 pm the day it is due.

Office Hours: Please use this link to tell me your best guess at your availability this quarter www.when2meet.com/?35843515-TB7hy.

My office hours are drop-in and available to all students. Typically, students come to my office hours to discuss questions from recent assignments, exceptions to requirements (e.g. late written work submission), current grade, or concerns about the course. That said, any discussion is welcome! I will make private time available to those who wish to speak about a personal matter, but I will also encourage cooperation with students present if working on common questions from the class.

If you are unable to attend my standard office hours, I also schedule individual appointments (typically in 20-minute blocks to accommodate several appointments each week).

Instructor Communication:

Our class will communicate through our Canvas site using announcements. If you want announcements show up in your UO email, check and adjust your settings under Account > Notifications.

Attendance:

- **Lectures Monday-Wednesday** Attendance is strongly recommended. If you miss class, you are still responsible for the material covered and any worksheets completed during classtime.
- **Lectures Thursday:** Attendance is required. Quizzes & exams will occur during these class sessions. Makeup policies for these absences listed on page 2.

Expected Classroom Behavior:

Students are expected to behave respectfully toward each other and toward the instructor during class time. This includes refraining from using cell phones during lectures.

University Policies

Instructors are now directed to refer students to the following link for information about general university policies, such as incompletes, academic accommodations, discrimination resources, etc: teaching.uoregon.edu/university-course-policies.

Prerequisite: C- or better in Math 252. The quiz on Thursday of week 1 (April 3rd) will cover this content.

Workload: A prepared student can expect to spend an average of 12 hours per week on this course. A student with weak pre-requisite skills should anticipate needing additional time to master the course material.

Homework (5% of final grade):

Homework will be assigned each week and due at the start of class on **Wednesdays**. This will be your primary means of feedback apart from quizzes and exams. Problems will be posted on Canvas; please work through them and submit your work in person. I anticipate this primarily being done on paper, but if you prefer using a tablet etc you can email me a PDF of your work.

Your homework will be graded primarily on completion. Each week I will ask the papermarker to grade a few problems for correctness as well.

Late written homework submissions: There is no late written homework accepted. If you can't make it to class on the day homework is due, please upload a scan of your work to Canvas before 3:00 pm. Your lowest two written homework scores are dropped.

Reading Checks (5% of final grade):

I ask that you look through the content for the week *before* class to help prepare you for lecture. I will post a weekly reading check quiz on Canvas due at **11:59 pm on Sundays**. I only ask that you skim through the online notes, and questions will be graded generously. To help prepare my lecture I will also ask if you have questions about the material.

Late reading checks: There is no late work accepted. Your lowest two reading check scores will be dropped.

Worksheets (5% of final grade):

Frequently, class time will be spent working on in-class worksheets. Collaboration with your peers is recommended for in-class worksheets. These worksheets are collected at the end of class and graded for completion. Worksheets should be available on Canvas as well, so if you couldn't make it to class, please scan and upload your work to Canvas by 11:59pm on the day they are assigned.

Late worksheets: No late worksheets are accepted, but your lowest two worksheet scores will be dropped.

Quizzes (15% of final grade):

Most weeks there will be an in-class quiz, usually on Thursdays toward the end of class except on the day of an exam. These will be timed, 25-minute, pencil & paper quizzes given at the end of class. Quizzes are given so that both you and your instructor can gauge your grasp of recent material. The first quiz is on April 2nd and is a review quiz which covers the prerequisite material for the course.

Makeup quizzes: There will be no makeup quizzes, but your lowest two quiz grades will be dropped.

Midterms (20% of final grade each; 40% total): There will be two, pencil & paper midterm exams in this course. These will be timed, 50-minute exams that will be administered on Thursday, April 24th and Thursday, May 22nd. Equation sheets will be provided for both midterms as well as the final, and you are also allowed one 3 by 5 inch notecard, front and back.

Makeup exams: Apart from a few special cases outlined below, no makeup exams will be offered. At the end of the term, your lowest midterm score will be automatically replaced with your final exam score if that would benefit you.

Makeup Policies: Makeup exams and quizzes are only offered in a few specific situations: religious accommodation, accessible education, pregnant and parenting students, and students on military deployment. Please prioritize being in class each Thursday for quizzes and exams if you're able, but if for example you are sick, please stay home.

The course attendance policy is meant to cover ordinary, day-to-day absences. I understand that extraordinary circumstances happen. In the event of an such a circumstance, I will work with you to consider how we might make adjustments to course requirements where possible. Please let me know as soon as you are able if you experience an extraordinary event.

There is no need to explain or offer personal information about the nature of your emergency — I will trust that you only to activate this policy in an emergency situation.

Final Exam (30% of grade): The cumulative final exam is on:

Monday, June 8, 2026, 2:45 p.m..

This exam will be an in-person, paper & pencil exam. You will have 120 minutes for the exam. If you notice a conflict *with a final for a different course*, please let me know as soon as possible to ensure that your conflict is accommodated. Otherwise, the final exam date is not negotiable and cannot be rescheduled.

The final exam is cumulative, but will emphasize newer content.

Grading: The course grade will be a weighted average of your scores on the course work as follows:

Category	Portion of Course Grade
Worksheets	5%
Written Homework	5%
Reading Checks	5%
Quizzes	15%
Exam 1	20%
Exam 2	20%
Final Exam	30%

Grading Scale: Your letter grade is determined by your final percentage:

A+	98–100%	B+	88–89%	C+	78–79%	D+	68–69%	F	0–59%
A	92–97%	B	82–87%	C	72–77%	D	62–67%		
A–	90–91%	B–	80–81%	C–	70–71%	D–	60–61%		

I don't intend on curving grades at the end of the quarter, but I will round your grade up to the next whole number (e.g. 89.01% is an A-).

Learning Outcomes: Math 253 begins with the study of sequences and series, and determining convergence theorems. The main goal is applying Taylor's theorem and understanding standard Taylor polynomial approximations like those for $\sin(x)$, $\cos(x)$, e^x and $\log(x)$. A successful student can...

- Decide if a given sequence converges or not.
- Express an indicated sum using Σ notation in closed form.
- Compute partial sums and other finite sums.
- State the precise definition of what it means for a sequence to have a limit.
- State the precise definition of what it means for a series to converge.
- Decide if a given series converges or not, using the Comparison Test, Divergence Test, Root Test, Ratio Test, Integral Test, Limit Comparison Test, Alternating Series Test, or a combination thereof, as appropriate.
- Decide if a given series converges or not using the definition.
- Evaluate the Taylor polynomial for a given function, given a center and a degree, by computing derivatives.
- Compute the Taylor polynomial for a rational function by performing long division.
- Use Taylor polynomials to approximate the values of functions.
- Given an easy sequence that converges to a limit L , together with an ϵ , determine an N such that $|a_n - L| < \epsilon$ for $n \geq N$.
- Given an alternating series and an epsilon, determine how many terms are needed to have the partial sum within epsilon of the limit.
- Find the interval of convergence of a given power series.
- Determine if a given series is absolutely convergent.
- Given a function, a center a , a degree d , and an accuracy level ϵ , determine an interval about a for which the d th Taylor polynomial is within ϵ of the function at all points.
- Use Taylor's Inequality to bound the error of a Taylor approximation.
- Differentiate and integrate functions given by power series.
- Given a differential equation, find the general solution as a power series up through a given degree. Also find particular solutions.
- Answer basic conceptual questions involving convergence of sequences and series, and also give examples of related phenomena.

Tentative Course Schedule: I have included below a rough schedule we will be following this term. The exam dates are fixed but the speed at which we move through a particular section may vary from the schedule.

Week	Sections Covered	Important Dates
1	Section 4.1: Sequences	Quiz 1 (Review)
2	Section 4.2: Series	Quiz 2
3	Section 4.3: Divergence and Integral Tests	Quiz 3
4	Section 4.4: Comparison Tests	4/24 – Midterm 1
5	Section 4.5: Alternating Series	Quiz 4
6	Section 4.6: Ratio and Root Tests	Quiz 5
7	Section 4.7: Power Series	Quiz 6
8	Section 4.8: Properties of Power Series	5/22 – Midterm 2
9	Section 4.9: Taylor Series	Quiz 7; No class – Memorial Day 5/25
10	Section 4.10: Applications of Taylor Series, Catch-up, Review	Quiz 8

Important Dates:

Saturday, 4/4 – last day to drop without a ‘W’

Monday, 4/6 – last day to add a class

Monday, 5/25 – No class – Memorial Day

Friday, 6/3 – last day to withdraw from the course or change the grading option to P/N

Learning Environment: The University of Oregon is working to create inclusive learning environments. Please notify your instructor if there are aspects of the instruction or design of this course that result in disability-related barriers to your participation. You are also encouraged to contact the Accessible Education Center at 541-346-1155 or uoaec@uoregon.edu. For those of you who are currently registered with Accessible Education Center for a documented disability, please present your paperwork to your instructor early in the term so that we can design a plan for you.

Student Conduct:

Your instructor plans to treat every student with respect and, as such, expects students to show respect for me and for the class as a whole. Violations of the student conduct code results in the incident being included on your student conduct record as well as academic sanctions such as a failing grade on any coursework related to the violation or simply a failing grade in the course. The University of Oregon requires all instances of cheating be reported, no matter how small. Cheating includes, but is not limited to:

- Looking at another student’s exam during a test
- Copying the work of another person (student, tutor, internet resource or otherwise) and submitting it as your own
- Using any materials except those explicitly approved during a test-taking situation
- Resubmitting graded work that was altered after being returned
- Cooperating on work for the course (including exams) without being explicitly allowed to do so
- Making solutions to homework, quizzes, exams, and other assignments available to anyone else. This includes solutions written by your instructor.

For a list of other descriptions of cheating, see the [Student Conduct Code](#). **Use of AI (like ChatGPT):**

There are responsible uses for AI. Asking it to critique a solution, or a hint for how to begin a tough problem, or suggestions for next steps might be responsible uses of AI (the same way that you might ask your math major roommate or a tutor to help out). However, there are many unacceptable uses of AI constituting academic dishonesty, such as:

- Consulting AI during a test
- Asking it how to solve a homework problem and passing the work off as your own
- Submitting an answer produced by AI without understanding what it means

AI seems to be here to stay and instructors around the world, like it or not, are coming to terms with that. It can be ubiquitous as a study tool if you want a source of additional homework problems, but there's the obvious temptation of relying on it to do your work for you. AI will not (and should not) be available during quizzes and exams, so it is your responsibility to determine what constitutes an appropriate use of AI to best prepare you for success in this course.

Prohibited Discrimination and Harassment Reporting I am a student-directed employee. For information about my reporting obligations as an employee, please see [Employee Reporting Obligations](#). Students experiencing any form of prohibited discrimination or harassment, including sex or gender based violence, may seek information on [safe.uoregon.edu](#), [respect.uoregon.edu](#), [titleix.uoregon.edu](#), or [aaeo.uoregon.edu](#) or contact the non-confidential Title IX office (541-346-8136), AAEO office (541-346-3123), or Dean of Students offices (541-346-3216), or call the 24-7 hotline 541-346-SAFE for help. I am also a mandatory reporter of child abuse. Please find more information at [Mandatory Reporting of Child Abuse and Neglect](#).

General Suggestions:

- Don't get behind in your homework, start it when it is assigned so there is time to ask questions.
- Participate in class, ask questions, and make use of office hours.
- Read the text! Frequently, the in-class examples will differ from those in the text.
- Work together! Feel free to work on homework together, but each student must complete their own work.
- Look over your graded work to be sure that you understand your mistakes.
- Keep all your old exams, homework, and quizzes to use when you're studying for tests.

Exams: Two midterm exams (weeks 4 and 8) will cover several sections of material and will be administered on **Thursdays** of the weeks listed below. The final exam will cover all sections, and will be administered in our regular classroom at 2:45 p.m. on Monday, June 8.

- 1) Midterm Exam 1: Thursday, Week 4 (April 24)
- 2) Midterm Exam 2: Thursday, Week 8 (May 22)
- 3) Final Exam: Monday, Week 11 (June 8)

Accessibility: For those of you who are currently registered with Accessible Education Center for a documented disability, please present your paperwork to me during the first week of the term (or earlier) so that we can design a plan for you. Those of you with a disability (or who might) but are not registered with AEC should contact them as soon as possible. It is much more likely that measures can be taken to provide adequate special accommodation if the organization is done through AEC. I have attempted to provide documents that are accessible. Please let me know if you need additional accommodations.

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- Using any materials except those explicitly approved during a test-taking situation
- Resubmitting graded work that was altered after being returned
- Cooperating on work for the course (including exams) without being explicitly allowed to do so

For a list of other descriptions of cheating, see the [Student Conduct Code](#).

Getting Help From...

Me: You should make use of my office time whenever possible. Keep in mind that I may ask you questions about how you started the problems and encourage you to contribute to solving it, rather than simply handing you the result.

Tutors: The Tutoring and Academic Engagement Center (Sky Studio on 4th floor of the Knight Library) has both free and private tutors available during most business hours.

Free tutoring is also available in the Math Library Reading Room (across the hall from the math office in Fenton Hall) on weekdays and Sundays.

Recommended First-week Review

We will start with discussing sequences. If differentiation or integration is a bit hazy, I'd recommend checking out Cruz's calculus notes.